piRover Builds with K2

Creating a Remote Connection Rev 2.0

Overview:

In this activity, you will connect to your Raspberry Pi and complete configuration steps so that you will have remote access to your piRover. You'll download and install the required software on your computer system and make your first remote connection into the system.

Prerequisites:

Prior to beginning the instruction provided in this lesson you must have completed the following:

1. piRover build and test

Performance Outcomes:

1. Connect to a Wi-Fi access point.
2. Edit a configuration.
3. Modify the Wi-Fi access point name and authentication.

Resources:

1. Yahboom G1 Tank Repository
2. RaspAP full-featured wireless router setup

Materials:

1. Wi-Fi access point
2. Laptop or Workstation with Ethernet network connection
3. Ethernet cable (provided in kit)
4. VNC Viewer
5. Bonjour Print Services for Windows

Directions:

1. A typical setup for a Raspberry Pi single-board computer is shown in Figure 1. Note the USB-C power connection in addition to the USB Dongle providing keyboard and mouse connectivity. A small HDMI monitor is displaying the Pi desktop and requires the Micro-HDMI cable included in your kit.

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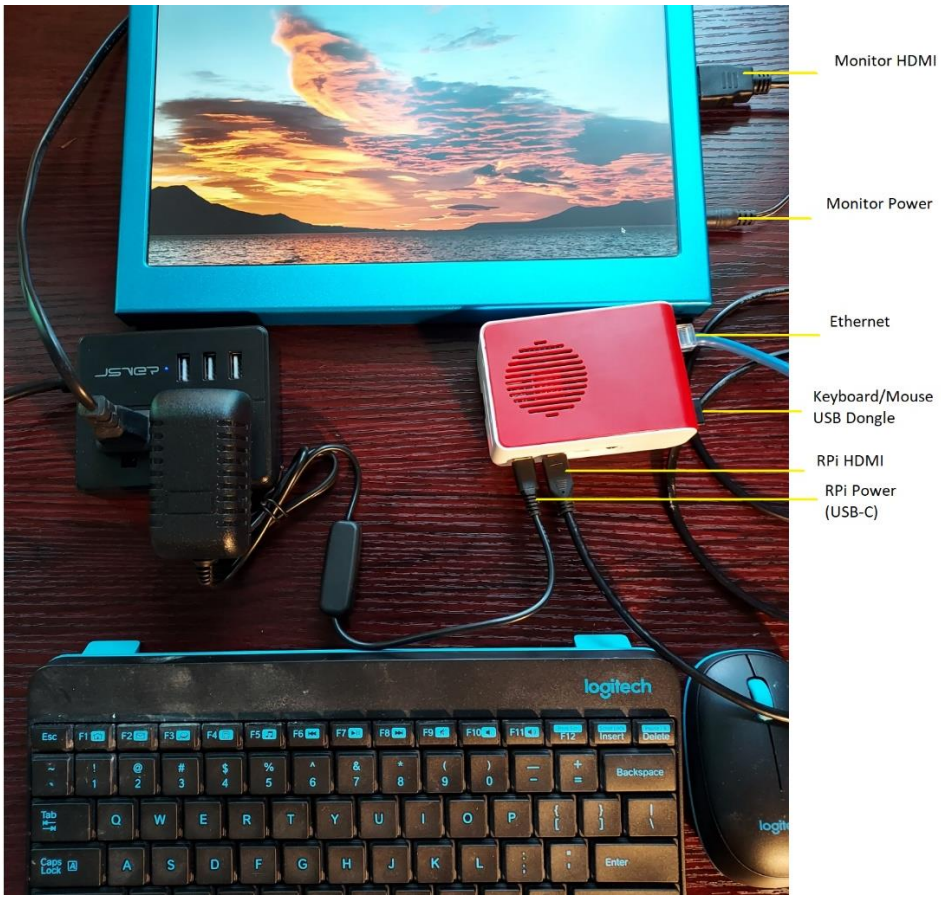
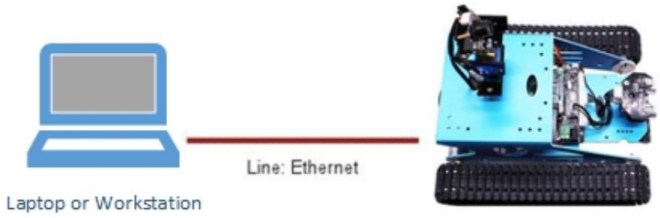


Figure 1 – RPi Single Board Computer Setup

2. In this class, you are connecting to the Raspberry Pi integrated into your piRover system. Typically, you will not have a monitor, keyboard, and mouse directly connected to the Pi, but these components may be required if you are having issues connecting. Normally you'll connect using a remote desktop service (VNC) running on the Pi. This is often referred to as "headless mode".

3. In this activity, you make your first connect to the RPi using the Ethernet cable provided. Eventually you may be able to connect using Wi-Fi connectivity and the Ethernet connection will not be required.

4. The first step in making the remote connection is to connect the piRover to your computer using the Ethernet cable provided.



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5. Recall that if your computer does not have an Ethernet port, a USB to Ethernet adapter is required like the one shown below. This requirement was discussed during the first week of class.

Figure 2 Ethernet to USB adapter

6. You'll be creating a peer-to-peer network connection between your workstation and the Raspberry Pi. If you're using a Windows PC an additional software driver is required to make this type of connection. If you are on a PC, navigate to this Bonjour Print Services for Windows link. Download and install the driver.

7. You'll use VNC remote desktop software to provide the remote connection. The RPi system provided for you has the VNC server feature enabled as shown below, but you must install the VNC client on your workstation.

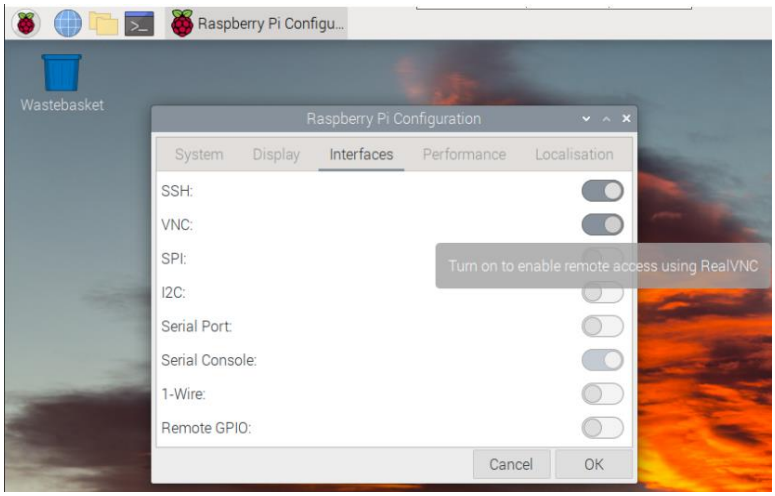

Figure 3 Remote Desktop showing VNC Server Enabled

6. Open the VNC Viewer link on your workstation. Download and install the VNC Viewer client software. **Just install the viewer software.** The software is free, and you DO NOT need to sign up for additional software or services such as VNC Connect.

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7. You're now ready to connect.
    a. Turn the power on to the piRover. Note the red LED indicating that power is applied and the flashing green LED light indicating activity.

    b. Open VNC on your workstation. The window below is shown.

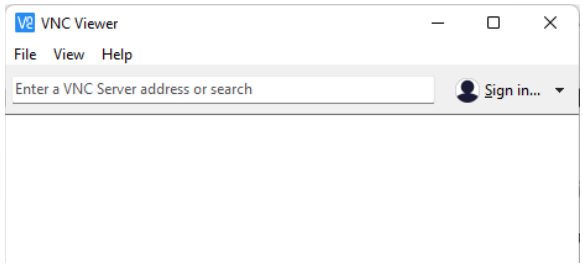

Figure 4 VNC Client Window

    c. Enter the VNC Server address below. The piRover host name is yahboomtank. The .local is appended to the end due to the peer-to-peer network connection created over the Ethernet.

    `yahboomtank.local`

    d. You must provide credentials to create the remote connection. Enter **pi** as the user and **yahboom** as the password. You'll continue to use pi as the user throughout the course. You'll change the password once you are connected.

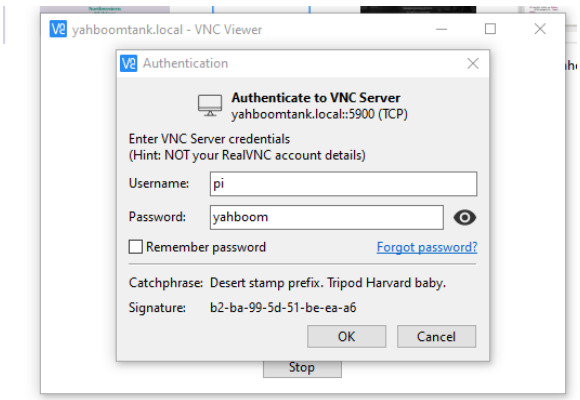

Figure 5 VNC Authentication

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e. Click OK. The remote desktop is displayed.

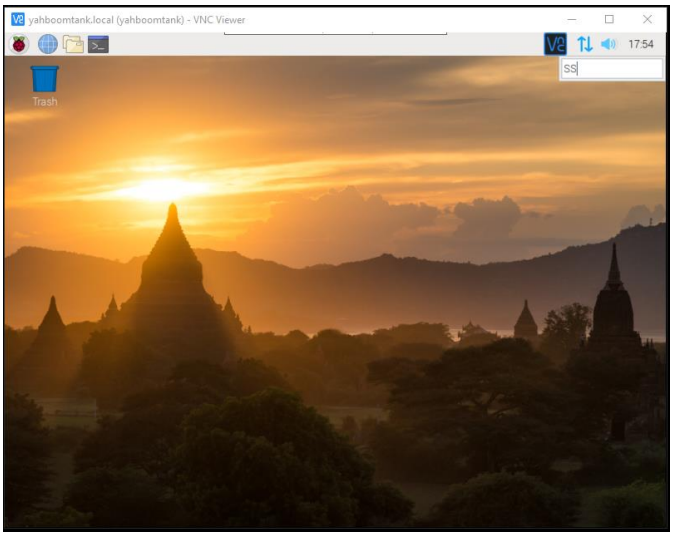

Figure 6 RPi Remote Desktop using VNC Viewer

8. Change the initial ***yahboom*** password to secure your system. Click the Raspberry in the top left corner and then click Preferences and then Raspberry Pi Configuration as shown below.

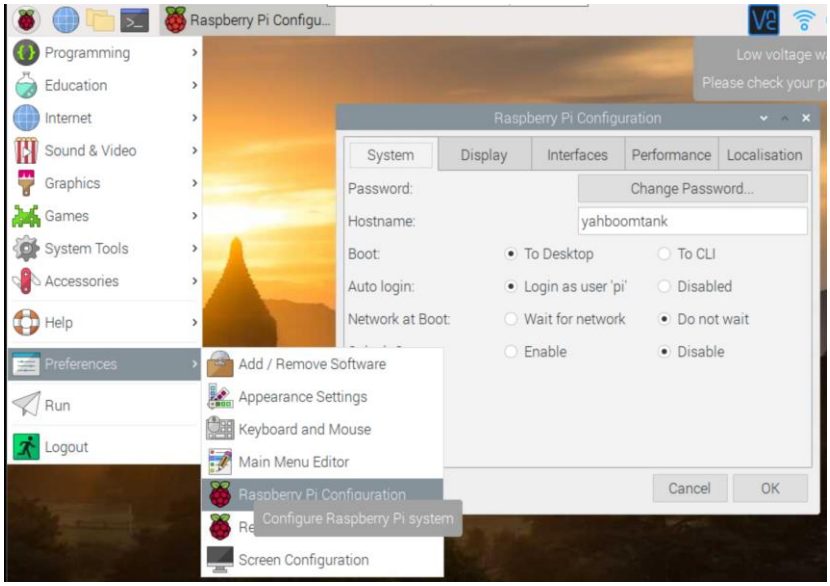

Figure 7 RPi Menu and Configuration

9. Click the Change Password button and enter your new password. Confirming the password will activate the OK button. Click OK and then close the configuration window. **NOTE:** *You must remember your password!* There is no administrative override or ability to reset. If you forget your password, you'll need to start with a new image of operating system and reconfigure.

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10. Now that the password is updated. Restart the piRover using the menu as shown below.

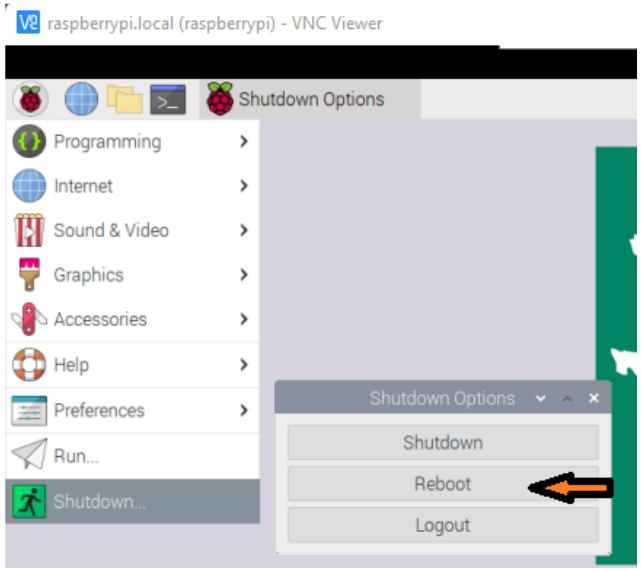

Figure 8 Reboot using the RPi Menu

11. The VNC view will lose the connection and attempt to reconnect.

12. Once the piRover boots, the Authentication screen will be displayed again. Enter your new password, click the *Remember password* check box, and click OK. The remote desktop will display.

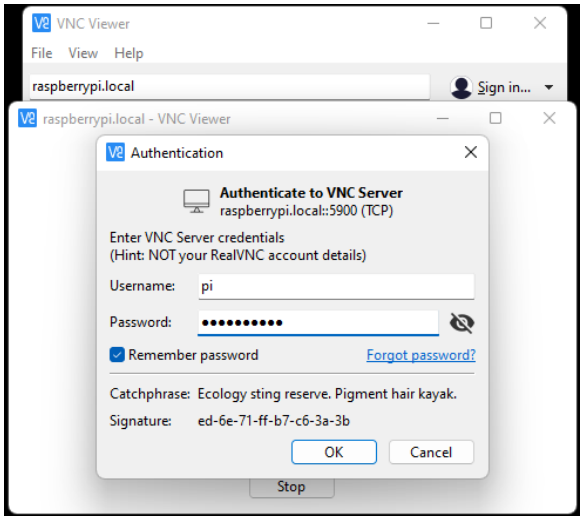

Figure 9 VNC Authentication with Remember Checkbox Ticked

13. The next step is to configure the piRover's Wi-Fi connection. Open the remote desktop and locate the Wi-Fi icon in the upper righthand location on the desktop.

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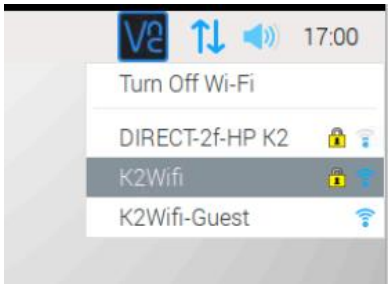
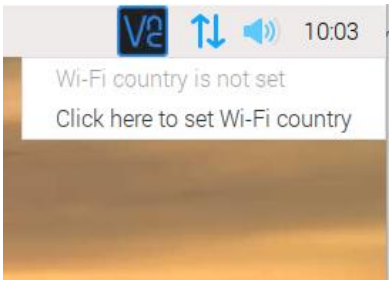
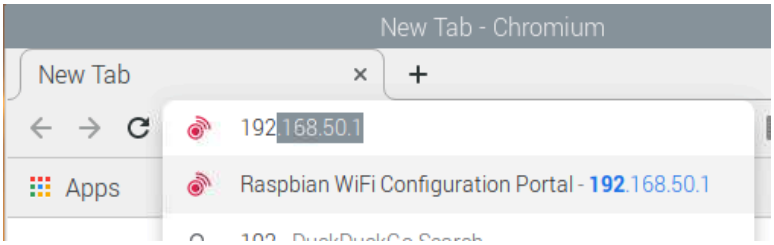
14. Normally, you click on this icon to select the Wi-Fi access point from the available list as shown in the image on the left. However, with this Raspian configuration you will likely see an error message related to the Wi-Fi country code not being set as shown in the image on the right.



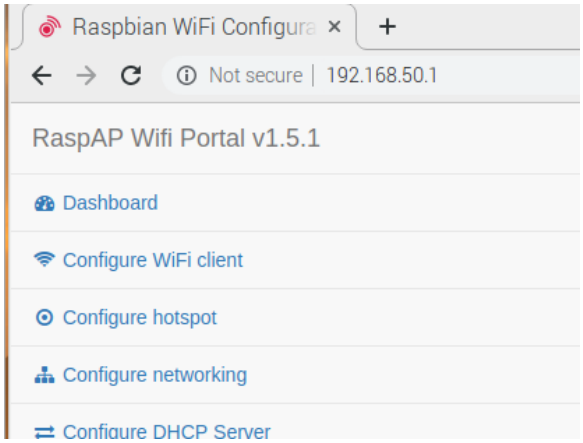


15. To connect to your Wi-Fi access point, open the browser and connect to the Raspian WiFi Configuration Portal at 192.168.50.1



16. This is a local site that requires authentication. The username is **admin** and the password is **yahboom**, but your system has stored these from prior access. You can just click Sign in.

17. Select the **Configure WiFi client** from the menu.



18. You will see the Wi-Fi access points stored on the system and also new connections available. You can delete the Yahboom connections likely used during factory setup. The image below shows the K2Wifi connections. You will see access points available to you. Connect to

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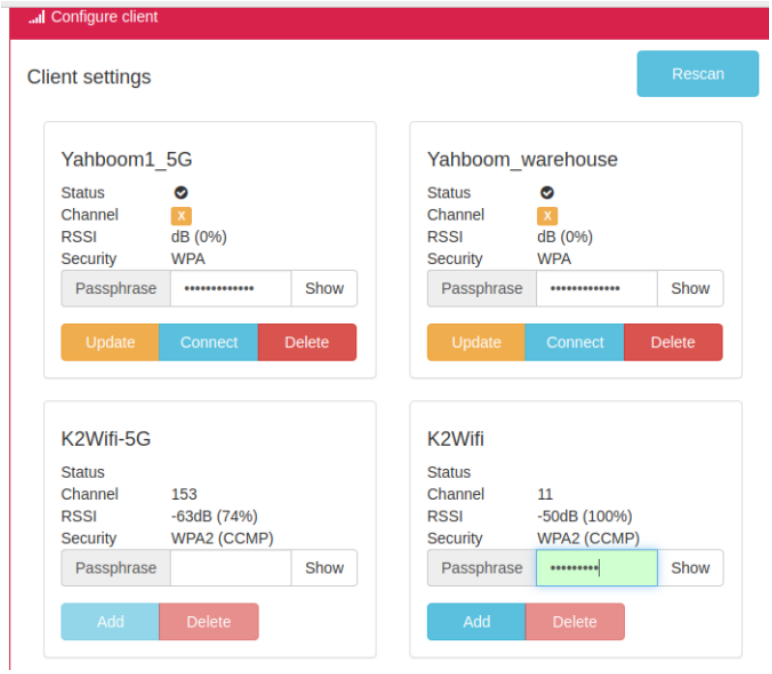
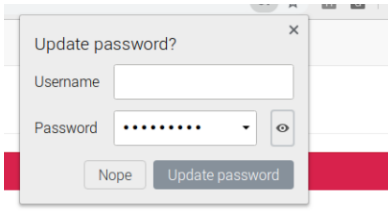
your home Wi-Fi or check with an instructor if you are connecting at school. A passphrase is required.



19. The browser prompts to save the password. Click Save.



20. The Wi-Fi client connection is made. Hover over the Wi-Fi icon to see the new IP address assigned to the wlan0 connection.

21. Note in the image above that there are two IP addresses shown. The uap0 connection is the piRover access point and is enabling the remote VNC connection from your workstation to the piRover. The wlan0 connection is the Wi-Fi client connection from the piRover to the home local area network. The uap0 IP address is same for all

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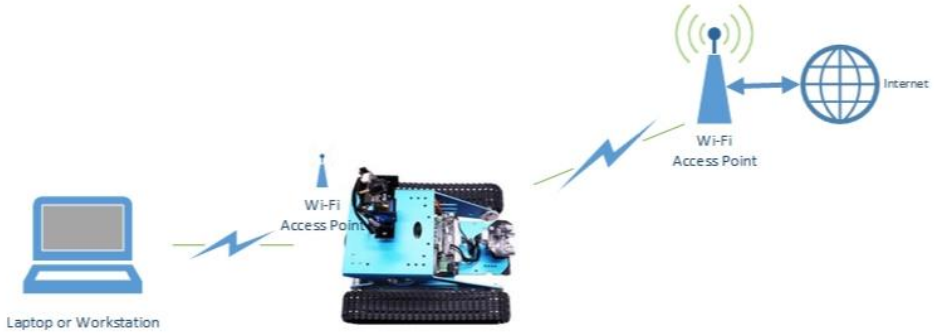
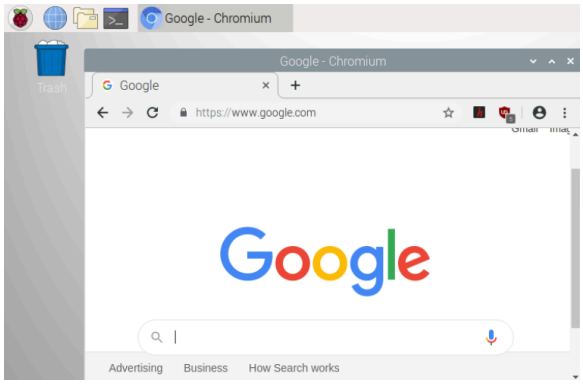
piRovers. Your wlan0 address will be different due to the unique connection to your local area network.

uap0: 192.168.50.1 wlan0: 192.168.1.10

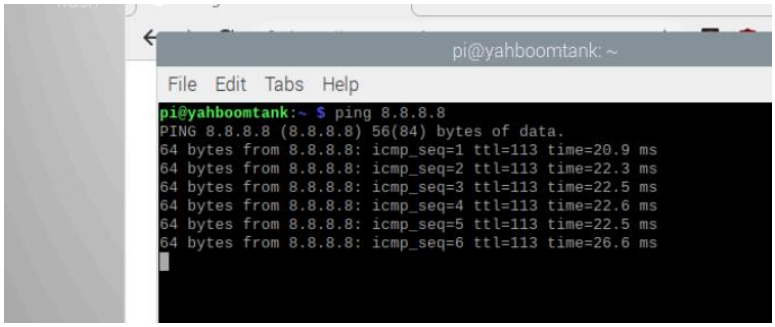


22. Verify that you have an Internet connection on the piRover by launching the browser using the icon on the taskbar and navigating to Google.com. The Google search page should load.



23. A more common way for a developer to check for an Internet connection is to "ping" a server. Open a terminal window and enter the following command. This "pings" the Google name space server at 8.8.8.8. If your connection is good, you will see replies as shown below. Close the window or use Ctrl+C to stop.

ping 8.8.8.8



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Assessment:

Follow along with the instructor to create a screen capture of your VNC remote connection to your Raspberry Pi. Name the file **VNC_remote.jpg**

Post this image to the assignment link provided. See the tools section for additional help on creating a screen capture. Submit your screen capture with other required files in your weekly submission at the bottom of this week's Moodle section.

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